Framework for Probability Samples Learning Activity

STA304 — Summer 2026

Learning Activity 6 Overview

In this activity you will apply statistical concepts from lecture 6 related to domain estimation.

Exercise

1. An SRS of 1,500 licensed boat owners in a state was sampled from a list of 400,000 names with currently licensed boats. Define two domains: domain 1 consists of the 472 licensed boat owners who own large motorboats (defined as open motorboats longer than 16 feet) and domain 2 consists of the 1,028 licensed boat owners who do not own large motorboats. The following frequency table gives the number of respondents in each domain who reported having that number of children. Estimate, along with a 95%

Number of Children	Domain 1	Domain 2
0	76	528
1	139	189
2	166	225
3	63	76
4	19	8
5	5	2
6	3	0
8	1	0
Total	472	1028

CI,

(a) The percentage of boat owners who have children in each domain.

Solution. Let $y_i = 1$ if household i has at least one child and $y_i = 0$ otherwise. The domain proportion estimator is $\hat{p}_d = \bar{y}_d = n_{d,\text{children}}/n_d$, and its estimated variance under SRS is

$$\widehat{\text{Var}}(\hat{p}_d) = \left(1 - \frac{n}{N}\right)\frac{n}{n-1}\frac{\hat{p}_d(1-\hat{p}_d)}{n_d}$$

Domain 1. Of $n_1 = 472$ respondents, $472 - 76 = 396$ have children, so

$$\hat{p}_1 = \frac{396}{472}$$

$$\widehat{\text{Var}}(\hat{p}_1) = 0.0002853256$$

$$\widehat{\mathrm{SE}}(\hat{p}_1) = 0.01689158$$

The 95% confidence interval is

$$\hat{p}_1 \pm t_{\alpha/2,n_1-1} \times \widehat{\mathrm{SE}}(\hat{p}_1) = (0.8057667,\ 0.8721752).$$

$$\boxed{\hat{p}_1 = 83.90\%, \quad 95\%\ \mathrm{CI} = (80.58\%,\ 87.22\%)}$$

Domain 2. Of $n_2 = 1028$ respondents, $1028 - 528 = 500$ have children, so

$$\hat{p}_2 = \frac{500}{1028}$$

$$\widehat{\mathrm{Var}}(\hat{p}_2) = 0.0002422605$$
$$\widehat{\mathrm{SE}}(\hat{p}_2) = 0.01556472$$

The 95% confidence interval is

$$\hat{p}_2 \pm t_{\alpha/2,n_2-1} \times \widehat{\mathrm{SE}}(\hat{p}_2) = (0.455839,\ 0.5169236).$$

$$\boxed{\hat{p}_2 = 48.64\%, \quad 95\%\ \mathrm{CI} = (45.58\%,\ 51.69\%)}$$

(b) The average number of children per household among boat owners in domain 1.

Solution. The domain mean estimator is

$$\bar{y}_1 = \frac{1}{n_1} \sum_{i \in \mathcal{S}_1} y_i,$$

with estimated variance

$$\widehat{\mathrm{Var}}(\bar{y}_1) = \left(1 - \frac{n}{N}\right) \frac{s_1^2}{n_1}.$$

Computing the sample mean directly from the frequency table:

$$\begin{aligned}
\bar{y}_1 &= \frac{0(76) + 1(139) + 2(166) + 3(63) + 4(19) + 5(5) + 6(3) + 7(0) + 8(1)}{472} \\
&= \frac{0 + 139 + 332 + 189 + 76 + 25 + 18 + 0 + 8}{472} = \frac{787}{472} = 1.667373.
\end{aligned}$$

The sample variance is

$$\begin{aligned}
s_{y1}^2 &= \frac{1}{n_1 - 1} \left(\sum_{i \in \mathcal{S}_1} y_i^2 - n_1 \bar{y}_1^2 \right) \\
&= \frac{1}{471} \left(0^2(76) + 1^2(139) + 2^2(166) + 3^2(63) + 4^2(19) + 5^2(5) + 6^2(3) + 8^2(1) - 472(1.66737)^2 \right) \\
&= 1.398678.
\end{aligned}$$

$$\widehat{\mathrm{Var}}(\bar{y}_1) = \left(1 - \frac{1500}{400000}\right) \frac{1500}{472^2} \frac{472-1}{1500-1} s_{y1}^2 = 0.0029479$$

$$\widehat{\mathrm{SE}}(\bar{y}_1) = \sqrt{0.0029479} = 0.05429457.$$

The 95% confidence interval is

$$\bar{y}_1 \pm t_{\alpha/2, n_1 - 1} \times \widehat{\mathrm{SE}}(\bar{y}_1) = (1.560683,\ 1.774062).$$

$$\boxed{\bar{y}_1 = 1.6674 \text{ children}, \quad 95\% \text{ CI} = (1.5609,\ 1.7741)}$$

2. The population count of licensed boat drivers who own a large boat is unknown here (N_1 is unknown using notation from class). Define estimators for the domain size and totals as:

$$\hat{N}_d = \frac{n_d}{n} N \qquad\qquad \hat{t}^*_{yd} = \hat{N}_d \bar{y}_d$$

(a) Estimate the total number of children who live in households where someone owns a large motorboat.

Solution. Using the given estimators with $d = 1$:

$$\hat{N}_1 = \frac{n_1}{n} N = \frac{472}{1500} \times 400{,}000 = 125{,}866.7$$

$$\boxed{\hat{t}^*_{y1} \approx 209{,}866.7 \text{ children.}}$$

(b) Let u_i be defined such that $u_i = y_i$ if $i \in \mathcal{U}_d$ and 0 otherwise. Show that

$$\hat{t}^*_{yd} = N \bar{u}_{\mathcal{S}}$$

where $\bar{u}_{\mathcal{S}}$ is the sample mean of u.

Proof.

$$\begin{aligned} N\bar{u}_{\mathcal{S}} &= N \cdot \frac{1}{n} \sum_{i \in \mathcal{S}} u_i = N \cdot \frac{1}{n} \sum_{i \in \mathcal{S}_d} y_i \quad \text{(since } u_i = 0 \text{ outside domain)} \\ &= N \cdot \frac{n_d}{n} \cdot \frac{1}{n_d} \sum_{i \in \mathcal{S}_d} y_i = \hat{N}_d\, \bar{y}_d = \hat{t}^*_{yd}. \quad \blacksquare \end{aligned}$$

(c) Justify that $\bar{u}_{\mathcal{S}}$ is sample mean estimator under SRS and write an expression for its estimated variance.

Solution. Under SRS, the sample $\mathcal{S}$ is drawn without replacement from all N population units. Because u_i is simply a function defined on every population unit (equal to y_i inside the domain and 0 outside), the sample values $\{u_i : i \in \mathcal{S}\}$ constitute an SRS of size n from the population $\{u_i : i \in \mathcal{U}\}$. Therefore $\bar{u}_{\mathcal{S}}$ is an unbiased estimator of the population mean $\bar{U} = \frac{1}{N}\sum_{i\in\mathcal{U}} u_i$ under the standard SRS framework, and its estimated variance is

$$\widehat{\text{Var}}(\bar{u}_{\mathcal{S}}) = \left(1 - \frac{n}{N}\right)\frac{s_u^2}{n},$$

where $s_u^2 = \dfrac{1}{n-1}\displaystyle\sum_{i\in\mathcal{S}}(u_i - \bar{u}_{\mathcal{S}})^2$ is the sample variance of the u_i values.

(d) Write an expression for the standard error of $\hat{t}^*_{yd}$.

Solution. Since $\hat{t}^*_{yd} = N\bar{u}_{\mathcal{S}}$,

$$\widehat{\text{Var}}(\hat{t}^*_{yd}) = N^2\,\widehat{\text{Var}}(\bar{u}_{\mathcal{S}}) = N^2\left(1 - \frac{n}{N}\right)\frac{s_u^2}{n}.$$

Therefore the estimated standard error is

$$\boxed{\widehat{\text{SE}}(\hat{t}^*_{yd}) = N\sqrt{\left(1 - \frac{n}{N}\right)\frac{s_u^2}{n}}.}$$

(e) Discuss whether you **have enough information** to calculate the standard error you found in (d).

Solution. Yes, we have enough information. The expression

$$\widehat{\text{SE}}(\hat{t}^*_{yd}) = N\sqrt{\left(1 - \frac{n}{N}\right)\frac{s_u^2}{n}}$$

requires only $N = 400{,}000$, $n = 1{,}500$, and the sample variance

$$s_u^2 = \frac{1}{n-1}\sum_{i\in\mathcal{S}}(u_i - \bar{u}_{\mathcal{S}})^2.$$

All of these are known: N and n are given, and s_u^2 can be computed directly from the observed data because we know which sample units belong to domain 1 (and have their y_i values) and which belong to domain 2 (for which $u_i = 0$).

For completeness, using our data:

$$s_u^2 = \frac{1}{1499}\sum_{i\in\mathcal{S}}(u_i - \bar{u}_{\mathcal{S}})^2 = 1.03942,$$

giving

$$\widehat{\text{SE}}(\hat{t}^*_{y1}) = 400{,}000\sqrt{(0.99625)\frac{1.03942}{1500}} \approx \mathbf{10{,}509.78} \text{ children.}$$